

Light: Unit Review Homework

Year 9 Science · about 45 minutes

Name: _____

Use a ruler for every ray; put arrows on rays and draw virtual rays dotted. In Section B, explain in full sentences: say what the light does, why, and what we see as a result. The level each question is aimed at is shown on the right.

Section A

about 20 minutes

A1 Write **S** (light source) or **R** (light reflector) for each.

[A]

the Sun _____ the Moon _____ a candle flame _____

a bicycle reflector _____ a mirror _____ a glow-worm _____

A2 Is each material transparent, translucent or opaque?

[A]

frosted glass _____ clear plastic bottle _____

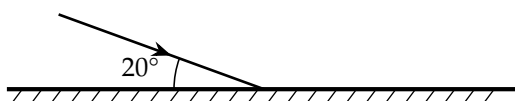
aluminium foil _____ tracing paper _____

A3 The Moon is 384 000 km from the Earth. Light travels at 300 000 km/s. Calculate how long light takes to travel from the Moon to the Earth. Show your working.

[A]

A4 A ray hits a plane mirror at 20° to the mirror surface.

[A]



(a) Draw the normal and the reflected ray.

(b) Angle of incidence = _____ Angle of reflection = _____

A5 A student stands 2.5 m in front of a plane mirror.

[A]

(a) How far is the image from the mirror? _____

(b) The student walks 1.0 m towards the mirror. How far is the student from their image now?

A6 Complete the table.

[A]

Object	Light shone on it	Colour it appears
green leaf	white light	
white T-shirt	blue light	
red rose	green light	

A7 For each ray, write *towards the normal*, *away from the normal* or *no bending*.

[A]

- (a) From air into water, at an angle _____
- (b) From glass into air, at an angle _____
- (c) From water into air, along the normal _____

A8 Name the part of the eye.

[A]

- (a) the coloured muscle that controls the size of the pupil _____
- (b) the layer of light-sensitive cells _____
- (c) carries signals to the brain _____
- (d) the clear front of the eye, which refracts light the most _____

Section B

about 25 minutes

B1 A yellow flower is lit with blue light in a dark room. Describe and explain the colour it appears.

[M]

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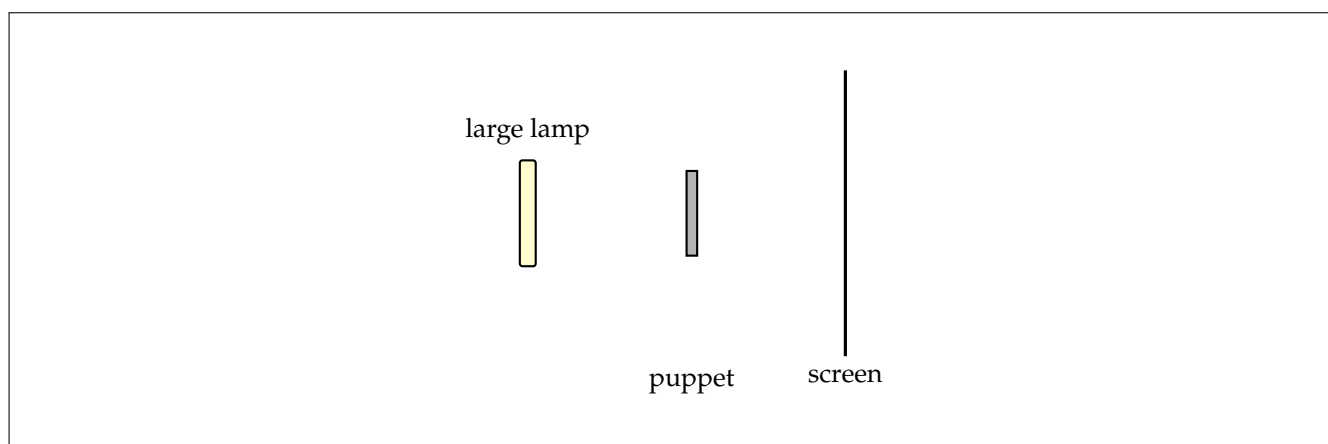
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B2 A shadow puppet is held between a large lamp and a screen.

[M/E]



- (a) Draw rays from the top and bottom of the lamp past the puppet to the screen. Shade and label the umbra and the penumbra.
- (b) Explain why the edges of the shadow are fuzzy.

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(c) The puppet is moved closer to the lamp. Explain what happens to the size of the shadow.

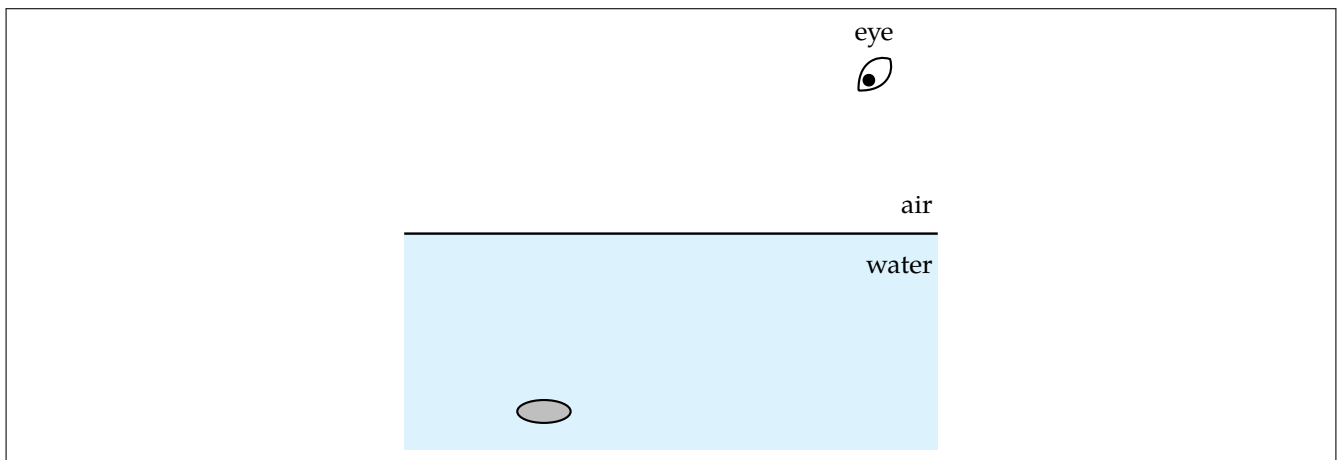
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B3 A coin lies on the bottom of a swimming pool.

[M/E]



- (a) Draw a ray of light from the coin to the eye. Show where the coin appears to be.
- (b) Explain why the coin appears closer to the surface than it really is.
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B4 Explain how a normal eye forms a clear image of a distant object. Then explain why a short-sighted person sees distant objects blurred, and how a concave lens helps. **[E]**

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B5 A dancer wears a white shirt, green trousers and a blue hat. A spotlight shines white light through a magenta filter onto the dancer. Describe and explain the colour of each item that the audience sees. **[E]**

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B6 Explain why a prism splits white light into a spectrum, and why violet light ends up on the opposite side to red light. **[M/E]**

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